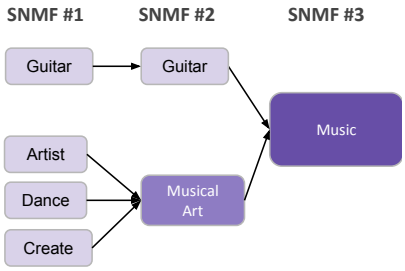

Figure 11: Merging of Music related MLP features.

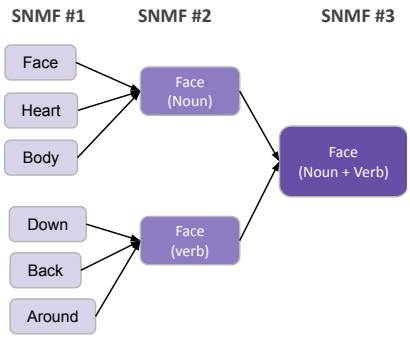

Figure 12: Hierarchy showing that semantically different, but syntactically related features exhibit merging as well.

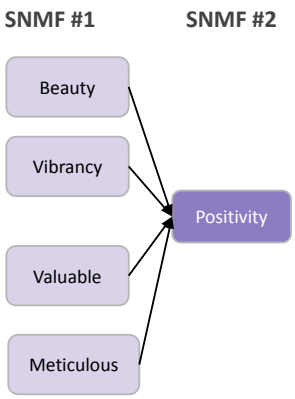

Figure 13: Hierarchy showing that semantically related features share neurons.

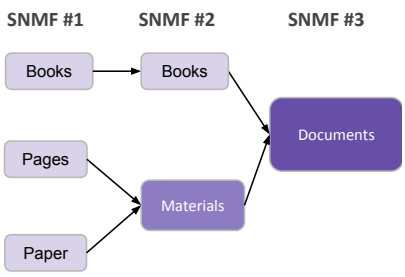

Figure 14: Hierarchy showing that compositional concepts exhibit neuron sharing.

G Concept Detection Prompts

In this section we present the prompts used for the concept detection experiments.

G.1 Generating Dataset From Concept Descriptions

The following prompt is used sequentially to generate the dataset employed in identifying MLP features with SNMF in Section 4 and Section 5. To generate the dataset, we provide the concept and prompt the model m times, where m is the desired number of sentences per concept.

Dataset Generation Prompt

Generate a sentence that exemplifies and strongly exhibits the concept of `{concept}`.
Make sure the sentence is different from the following list: `{current_sentences}`.
Output the sentence **ONLY** without additional text.

G.2 Generating Activating and Neutral Sentences

The following prompts are used to generate sentences that exemplify a concept (activating sentences) and those that do not (neutral sentences). To construct the list, we sequentially prompt the model to generate a new sentence while conditioning on the current list to encourage diversity.

Activating Prompt

You are given a description of an LLM concept.
Given the description, generate a sentence that contains tokens that would activate the feature.
Make sure your generated sentence exemplifies all the key terms and structures specified in the description.
Ensure that the sentence is a full, grammatically correct English sentence.
Category description:
`{description}`
Make sure you generate a sentence that is distinct from: `{previous_sentences}`

Output only the sentence without any additional text.

Neutral Prompt

You are given a description of an LLM concept.
Your objective is to generate a neutral sentence that should not activate the feature.
This means that you must not include in the sentence any tokens that relate to the feature.
Have the generated sentence be on a completely unrelated topic.
Concept description:
`{description}`
Make sure you generate a sentence that is distinct from: `{previous_sentences}`

Output only the sentence without any additional text.
Make sure to exclude any tokens that may activate the concept.

G.3 Input Description Prompt

The following prompt is used to generate a natural language description for each concept, based on the top activated tokens in the matrix Y, which represents token-level activations for each MLP feature. Specifically, for a given MLP feature, we extract the tokens with the top m highest activations and use them as input to the prompt. The model is asked to interpret the shared semantic content of these tokens. We then parse the *Results* section of the model's response to retrieve the final concept description.

Input Description Prompt

You are given a set of tokens, their surrounding context (words before and after the token), and an importance score.
Your task is to determine what is the connection between all the tokens.
Instructions:

1. **Focus on High-Importance Samples:**
   Examine only the token-context pairs with the highest importance scores. If a significant drop is observed beyond a threshold, ignore the lower-scoring pairs.

2. **Assess Token Consistency vs. Contextual Patterns:**
   - **Token Consistency:** If the high-importance samples are mostly identical tokens or strongly related tokens, then consider the tokens only as the primary contributor.
   - **Contextual Patterns:** If the tokens are not related to one another, then focus on common semantic, syntactic, or structural patterns in the surrounding contexts.

3. Always choose the simplest and most obvious underlying connection. If inspecting the tokens alone is enough to find a connection, do not mention or utilize the contexts.

Output Format:

- **Analysis:** `<reason what the underlying connection is>`
- **Results:** `<Single sentence description of the single most obvious connection between the tokens>`

Input:
`Token-Context Pairs: {token_context_str}`
Remember, find the most obvious connection prioritizing connecting the tokens alone without the contexts and only if you cannot find any connection between the tokens then you may inspect their contexts.